# Auditing Selection Purity and Transfer Utility under Joint Model and Data Heterogeneity

**Anonymous submission**


## Abstract

A canonical probability interface makes cross-architecture exchange well-defined across language models with different parameters and tokenizers, but it does not ensure that cleaner selected client data yields more useful server supervision. We introduce a four-arm audit that separates clean-transfer utility, contamination sensitivity, and the complete routing intervention, together with matched-size comparisons that probe selection order separately from subset size. A post-primary reused-test TREC decomposition first exposes opposing effects: severe uniform transfer exceeds clean transfer by $1.58\,pp$, whereas routing reduces severe-uniform accuracy by $1.42\,pp$, despite raising audited clean fraction and lowering descriptive score NLL and score ECE. This discovery motivates a role reversal frozen before model evaluation on the 25,000-review IMDb test, with one ten-seed primary contrast and a separate matched-$K$ secondary family. AG News supplies only prior task-specific context: routed transfer is $0.56\,pp$ below the anchor-only server in the original three seeds, and five later seeds retain the negative direction. The protocols use real checkpoints and hashed manifests; records are accepted only after complete-matrix verification, with paired uncertainty and explicit evidence roles. The contribution is diagnostic rather than algorithmic: selection purity, canonical score quality, and downstream transfer utility must be evaluated as distinct endpoints.


## Introduction

Federated fine-tuning promises to adapt language models without moving private client text. Two assumptions make this problem look easier than it is. First, many methods assume compatible client parameters even though deployed models can differ in width, tokenizer, and architecture. Second, evaluations often assume that a participating client's data is uniformly useful for the server task. Real client corpora can instead contain a mixture of useful examples, unrelated tasks, and responses that directly conflict with the target capability.

These axes interact. Parameter-space averaging may be undefined across model families, but moving to logits or probabilities only solves the interface problem; it does not establish whether transferred signals optimize the target server metric. Conversely, a local data filter that works for one model family may exploit architecture-specific loss scales and fail on another. We ask: *when client architectures differ, does raising audited clean fraction improve the utility of probabilities transferred to the server?*

We make three contributions. First, we formalize a four-arm audit for heterogeneous client-to-server distillation: public-anchor only (A), clean uniform transfer (B), severely contaminated uniform transfer (C), and severely contaminated routed transfer (D). The adjacent contrasts B–A, C–B, and D–C prevent a near-zero endpoint from hiding opposing transfer effects. Second, we separate discovery from prospective testing: a reused-test TREC decomposition generates the hypothesis, an IMDb test not previously used for model evaluation in the retained workspace supplies the single prospective cross-task test, and AG News supplies prior task-specific evidence. Third, fixed-size random, routed, and maximum-attainable-purity comparisons probe subset size, selection order, and audited composition. Content-addressed protocols and complete run records preserve these inferential roles. This is an evaluation contribution, not a new optimizer, privacy mechanism, or claim that purity and utility are universally unrelated. Figure 1 summarizes the estimands and inferential roles.

## Related Work

**Federated PEFT and model heterogeneity.** FedAvg aggregates compatible client parameters (McMahan et al. 2017), while FedMD and FedDF use public-data distillation to relax the shared-architecture assumption (Li and Wang 2019; Lin et al. 2020). LoRA makes adaptation economical by training low-rank updates (Hu et al. 2022), but factor exchange does not by itself resolve architectural incompatibility. FedBiOT compresses a server LLM for client-side adaptation (Wu et al. 2024). HetLoRA supports heterogeneous client LoRA ranks (Cho et al. 2024), while FLoRA avoids cross terms through stacking (Wang et al. 2024). FedMKT transfers knowledge between server LLMs and client SLMs (Fan et al. 2025). These works establish that heterogeneous LoRA and cross-size knowledge transfer are not new. Selective-FD filters misleading client predictions and ambiguous ensembles (Shao, Wu, and Zhang 2024); FedDFA prioritizes decision-boundary information (Wang et al.

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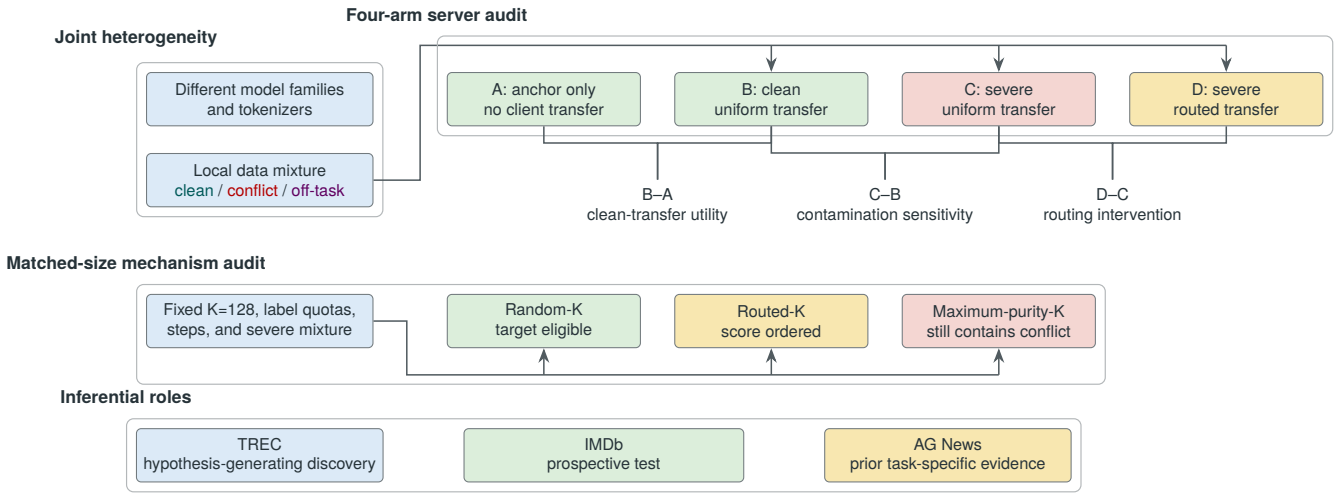



Figure 1: Evidence-aware four-arm audit. A–D separate clean-transfer utility, contamination sensitivity, and the complete routing intervention. The matched-size audit holds $K$, claimed-label quotas, optimizer steps, and the severe mixture fixed. TREC is hypothesis-generating discovery, IMDb supplies the prospective test, and AG News supplies prior task-specific evidence.

2025); and heterogeneous-distillation temperature has been reconsidered explicitly (Qi et al. 2025). Most directly, FedFD shows that heterogeneous logit ensembles can be unstable and proposes aligned feature distillation (Li et al. 2025). We therefore do not claim that logit transfer failure is new. Our narrower contribution audits decoder-only LMs whose clients are also internally contaminated, separating training-example selection, canonical probability transfer, and server utility. Our severe-unfiltered arm is a FedMD/FedDF-style uniform canonical-probability ensemble-distillation control adapted to decoder-only verbalizer classes; it is not an exact reproduction of either algorithm.

**Noisy labels and data selection.** The early-learning behavior of deep networks motivates small-loss selection (Arpit et al. 2017). Co-Teaching exchanges peer-selected small-loss examples (Han et al. 2018); trusted clean data can also estimate noise corrections (Hendrycks et al. 2018). Superfiltering shows that a weak language model can select instruction data for a stronger model (Li et al. 2024). Delora decouples a dual-LoRA noisy-label detector from downstream PEFT (Yuan, Chen, and Zhang 2025), and FedClean performs clean-sample selection, multi-stage finite-class relabeling, and robust aggregation (Jiang and Zhang 2025); FedNed instead addresses extremely noisy clients by negative distillation (Lu et al. 2024). Targeted instruction selection such as LESS optimizes downstream influence rather than label purity (Xia et al. 2024). Thus purity–utility separation is not a general insight we originate. Our public-anchor and gradient scores are diagnostic interventions used to test whether higher audited purity predicts server or local utility in frozen heterogeneous-LM protocols; we do not claim to outperform these methods in their native settings.

## Problem Setting

Clients $c \in \{1, \dots, C\}$ own decoder-only models $f_c$, tokenizers $\tau_c$, and datasets $D_c$. The server model $f_s$ may have a different architecture and tokenizer. A client dataset is a mixture

$$D_c = D_c^{\text{clean}} \cup D_c^{\text{conflict}} \cup D_c^{\text{off}},$$

where conflicting examples retain a target-task input but provide an incorrect answer, and off-task examples use a different task interface. Clients and server share a small public target-task anchor set $A$.

For architecture-agnostic classification exchange, let $\mathcal{Y}$ be the canonical label set and $L = |\mathcal{Y}|$. Client $c$ transmits a probability vector over this label set for every anchor input. The client score vector contains token-normalized candidate-answer losses $\boldsymbol{\ell}_{ci}$, giving $q_{ci} = \text{softmax}(-\boldsymbol{\ell}_{ci})$ for anchor $x_i$. The server uses $\bar{q}_i = C^{-1} \sum_c q_{ci}$, computes its own canonical probabilities $p_s$, and minimizes $\mathcal{L}_s = \lambda_A \, \text{CE}(y_i, p_s) + \lambda_D \, \text{CE}(\bar{q}_i, p_s)$. The anchor-only control sets $\lambda_D = 0$; routed transfer uses the frozen $\lambda_D$. This interface is compatible by construction, but its utility depends on client data quality.

## Training-Only Trust Endpoints

### Class-conditional gradient routing

For a labeled example $z = (x, y)$, let $g(z)$ be the gradient of its causal answer loss with respect to local LoRA parameters and let $a_y$ be the mean gradient of public anchors from class $y$. Their agreement is

$$s_g(z) = \frac{g(z)^\top a_y}{\|g(z)\| \|a_y\|}.$$

Let $P(v_i) \in [0, 1]$ denote the average-tie percentile rank of value $v_i$ within client $c$, with lower values ranked first. For target-task examples, the inconsistency margin is $m(z) =$

$\ell_y(z) - \min_k \ell_k(z)$; off-task examples receive one plus the largest observed target-task margin. The risk is

$$R(z) = \tfrac{1}{3}\{P(-s_g(z)) + P(\ell(z)) + P(m(z))\}.$$

Because raw cosine scales differ by model family, the routing gate uses the relative statistic

$$r_c = \frac{Q_{0.25}(\{s_g(z) : z \in D_c\})}{\max[Q_{0.25}(\{s_g(a) : a \in A\}), 10^{-12}]}.$$

The gate triggers when $r_c < 0.8$; otherwise all examples are retained. A triggered client keeps the lowest-risk $\max(1, \mathrm{round}(0.5|D_c|/L))$ examples per claimed target class, with source ID breaking ties; off-task examples have no claimed class and are excluded. Zero gradients are handled by the clamped cosine denominator. Thresholds and quotas were chosen on development data and fixed before the corresponding official-test runs. Clean-versus-conflict annotations are audit-only and never enter the score. Off-task membership is known from the explicit task interface; such examples have no claimed target class and are excluded whenever the gate triggers.

## Evidence-Aware Experimental Design

Table 1: Evidence ledger. "Prospective" is relative to the retained workspace; hashes establish internal consistency, not external preregistration.

| Study | Design timing | Authorized role |
|---|---|---|
| TREC primary | Separately frozen, five seeds | Task-specific primary contrast |
| TREC four-arm | Post-primary, reused test | Discovery/mechanistic Holm family |
| IMDb four-arm | Frozen before official evaluation | Single prospective primary contrast |
| IMDb matched-$K$ | Post-development, pre-official | Secondary mechanistic Holm family |
| AG legacy | Three seeds plus post-observation check | Prior task-specific $D - A$ context |
| AG four-arm | Post-primary, run-disjoint rows | Secondary task-specific $D - C$ context |

Table 1 fixes the authorized inferential role of each study; shared hashes or benchmarks do not upgrade discovery or descriptive evidence into prospective confirmation.

**TREC discovery study.** TREC-Coarse provides a six-way answer-type task (Li and Roth 2002); IMDb remains the off-task source. Exact duplicate training lines are removed by raw-line hash before a balanced allocation of 72 anchors, 72 development questions, and three disjoint 120-question client pools. Each client class contributes 8 clean, 8 cyclic-conflict, and 4 off-task slots. The complete 500-question official test is evaluated over five seeds. The task, manifest, runner, full arguments, canonical $[72, 6]$ probability payloads, and serialized wire payloads are content-addressed. The model families, two rounds, update budget, and routing thresholds match the AG News study. The parent primary protocol was frozen after instrumentation repairs and before its official evaluation; those repairs and invalidated smoke runs are recorded in its protocol history. The later four-arm protocol explicitly records prior test access and its own audit-counter repair before any retained full aggregate.

**Prospective IMDb role reversal.** IMDb sentiment becomes the binary target and AG News the off-task source. A content-addressed manifest separates 128 public anchors, 1,000 development reviews, three disjoint 256-example client pools, and the complete 25,000-review official test. Each severe client contains 103 clean, 102 cyclic-conflict, and 51 off-task examples. Three development seeds authorize implementation only; no accuracy direction may tune or stop the complete ten-seed official matrix. The single primary contrast is D–C accuracy, using a two-sided paired Student-$t$ test and 95% seed interval. The narrow negative-direction gate also requires routing to raise the audited clean fraction above 103/256. The test was not decoded for model assessment until the protocol, development matrix, verifier, and analysis rule had been frozen.

After development but before official-test access, we froze a separate matched-size secondary addendum. Random-K, routed-K, and maximum-purity-K all select $K = 128$ with 64 examples per claimed label and share the severe mixture, optimizer steps, and server settings. Maximum-purity-K contains all 103 clean examples plus 25 conflicts and is therefore not an oracle-clean arm. Routed-minus-random and maximum-purity-minus-routed accuracy form exactly one two-sided Holm family; they cannot alter or confirm the original primary.

**AG News task-specific context.** AG News is the four-way target task and IMDb supplies real off-task text (Zhang, Zhao, and LeCun 2015; Maas et al. 2011). Conflicts use a deterministic non-identity label permutation. A frozen manifest separates 256 public anchors, 800 development examples, disjoint client pools, and the full 7,600-example official test. Clients are Qwen2.5-0.5B, SmolLM2-360M, and TinyLlama-1.1B; the server is Qwen2.5-1.5B. Its original three-seed comparison and five-seed same-test direction check precede a post-primary four-arm train-derived secondary study; each retains its own evidence label.

**Controls and statistics.** Within each server task, comparisons match server initialization, public anchors, server optimizer steps, prompts, and seed. The original AG News primary and seed replication execute shadow unfiltered clients but set $\lambda_D = 0$, so their recorded probability payload is unnecessary to the server rather than absent. The TREC anchor and the later AG four-arm runner omit the client path. Cost ratios are therefore protocol-specific and do not share one end-to-end work definition. Each run-level protocol is serialized before its corresponding execution; this does not make later phases globally preregistered once an earlier phase has accessed the same benchmark test. Paired Student-$t$ intervals characterize seed variation, not task- or sample-level uncertainty. TREC uses five designated primary seeds; its disclosed reused-test decomposition uses ten new common seeds and three adjacent two-sided accuracy contrasts in one

Holm family. IMDb uses ten common seeds, one original primary contrast, and the separate two-contrast secondary Holm family above. All other server metrics and net contrasts are descriptive. Raw records carry evidence labels, model and manifest hashes, split scope, wall time, and peak GPU allocation.

**Metrics and optimization.** For TREC, IMDb, and the new AG four-arm runner, class scores are negative token-normalized answer-suffix losses; we report accuracy, macro-F1, the true-label suffix loss, and 10-bin ECE from the resulting canonical scores. Because token-normalized suffix losses are renormalized across verbalizers, this is canonical verbalizer-score ECE, not calibration of the LM's native sequence probability. We additionally report normalized canonical class NLL where the runner records it. The legacy AG endpoint named `label_nll` instead records target-answer suffix loss and is not a canonical class NLL. All adapters target query/value projections with LoRA rank 4, scale 8, and no dropout, optimized by AdamW. The AG holdout uses two rounds of 40 client and 40 server steps (batch 8; server learning rate $10^{-4}$); IMDb matches this budget. Frozen JSON protocols contain the remaining length and selection settings. Peripheral local-adaptation protocols are detailed in the supplementary material.

## Results

### TREC discovery: a net comparison hides opposing effects

Across five primary seeds, routed-minus-anchor accuracy ranges from $-1.8$ to $+2.6\,pp$ with mixed signs. The mean is $+0.40\,pp$, with descriptive 95% seed interval $[-1.74, +2.54]\,pp$ and two-sided paired $p = 0.631$. Macro-F1 changes by $+0.73\,pp$ with an interval that also crosses zero. Canonical class NLL decreases by 0.0304 ($p = 0.0616$), while the secondary score ECE decreases by 0.0373 in every seed, with interval $[-0.0487, -0.0259]$ and nominal, multiplicity-unadjusted $p = 0.00081$. Thus TREC does not validate an accuracy benefit; the interval spans modest harm and benefit. Its lower score ECE is a descriptive secondary pattern. Routing triggers in 27/30 client-rounds and records 19,830 bytes of serialized probability tensors per run, excluding transport metadata. Its $9.61\times$ mean timed-region per-process wall-time ratio was measured with anchor runs serial and routed runs up to two-way concurrent; it is neither end-to-end nor an equal-work deployment benchmark.

After observing this compound comparison, we froze a post-primary four-arm ablation on the already accessed TREC test with ten new common seeds. The three adjacent accuracy contrasts form one two-sided Holm family. Clean uniform transfer versus anchor-only is inconclusive ($-0.20\,pp$, interval $[-0.71, +0.31]$, Holm $p = .397$). Contrary to a simple contamination-harm story, severe unfiltered transfer exceeds clean transfer by $1.58\,pp$ ($[+0.54, +2.62]$, Holm $p = .015$). The complete routed intervention then reduces severe-uniform accuracy by $1.42\,pp$ ($[-2.19, -0.65]$, Holm $p = .0074$), while descriptively reducing class NLL by 0.0314 and score ECE by 0.0596. The routed-minus-anchor net is $-0.04\,pp$ with interval $[-0.84, +0.76]$, without a fourth test. This same-test analysis diagnoses an accuracy–score-calibration trade-off; it is not a new holdout or independent confirmation and does not isolate which mixture component produces the severe-uniform gain. Routing triggers in 57/60 client-rounds and raises the selected clean fraction from 40.0% to 64.6% when off-task rows remain in the denominator, yet lowers accuracy relative to severe uniform transfer. Because this decomposition was designed after access to the same 500-example test, it motivates rather than confirms the cross-task hypothesis.

### AG News task-specific context

Across the original three seeds, routing loses $-0.56\,pp$ relative to the anchor-only server. The descriptive interval runs from $-0.76$ to $-0.36\,pp$, and every paired difference is negative. Routed distillation lowers answer loss and score ECE but costs $2.01\times$ the matched wall time. Accuracy and macro-F1 both decline; adding routed client targets is therefore harmful under AG News accuracy in this pipeline, not uniformly worse across predictive metrics. After observing this result, we froze five additional seeds without retuning. All five differences are negative, with mean $-0.80\,pp$; its descriptive interval runs from $-1.18$ to $-0.42\,pp$. The predeclared 4-of-5 directional heuristic passes, but it is not an alpha-controlled test and reuses the accessed official test. The combined eight-seed estimate ($-0.71\,pp$) is descriptive.

At development seed 42, severe AG News mixtures reduce local accuracy by 21.75–25.88 points across the three client families. Across Qwen seeds 42–44, routing improves severe local accuracy by $+3.37$ to $+30.00\,pp$, but this local improvement does not translate to the server endpoint. The detailed compatible-model and local-adaptation controls are reported in the supplementary material; they are not additional federated server endpoints.

## Discussion

Architecture compatibility and transfer utility answer different questions. Canonical probabilities create a common interface across model families, but their effect still depends on the target task, contamination, selection rule, and evaluation metric. The TREC decomposition illustrates why a routed-minus- anchor endpoint is insufficient: a near-zero net contrast can conceal opposing clean-transfer, contamination, and routing effects. Its lower descriptive score NLL and score ECE also cannot be substituted for accuracy utility.

The four-arm audit makes these effects identifiable at the intervention level. B–A asks whether clean client probabilities help beyond the public anchor; C–B asks what changes under the joint severe mixture; and D–C measures the complete routing intervention. The matched-size addendum then separates a change in selected-set size from selection order and attainable audited composition. This does not identify the contribution of each routing-score component, but it prevents purity enrichment from being credited when the comparison also changes $K$.

Inferential role is part of the result. TREC's reused-test decomposition is hypothesis-generating even with multiplicity control; the frozen IMDb role reversal supplies the prospective cross-task test; and the legacy AG News comparison is

prior task-specific evidence with a different $D - A$ estimand. Only the new AG four-arm study will provide a directly comparable $D - C$ contrast. Hashes and complete-matrix checks establish internal consistency, not external preregistration. The resulting claim is therefore an evaluation diagnosis rather than a new federated optimizer or a universal law relating purity and utility.

## Limitations and Ethics

The study simulates three clients from a fixed decoder-only model suite; it does not sample organizations, architectures, or tasks from a population. Conflict and off-task rows are controlled interventions, and the severe arm changes both simultaneously, so C–B cannot attribute an effect to either component. Seed intervals describe optimization variation rather than example- or task-population uncertainty. TREC's four-arm result reuses an accessed 500-example test, the IMDb matched-size family was designed after development, and AG's secondary rows are not an independent confirmation. Canonical verbalizer-score ECE is not native sequence-probability calibration. Pretraining contamination cannot be ruled out, and TREC's published license is recorded as unknown. Finally, logits, losses, anchors, and hashes provide no formal privacy; membership or reconstruction leakage is not measured. We make no privacy-preserving, production, or broad-generalization claim.

**AI-assistance disclosure.** OpenAI Codex assisted with code drafting, experiment orchestration, artifact validation, and manuscript editing. The human authors are responsible for the study design and submission, verified the cited sources and reported results against the raw artifacts, and take full responsibility for the paper's content.

## Reproducibility

The artifact contains content-addressed manifests and protocols, model-weight hashes, deterministic mixtures, per-run raw JSON, generated summaries, and reconstruction tests. Records are accepted only after complete-matrix verification of identity, data splits, arguments, protocol and runner digests, canonical probability payloads, selection composition, and evidence labels. Paper numbers and figures are regenerated from verified records rather than copied from console logs. The TREC primary and reused-test matrices, IMDb original and matched-size matrices, and their distinct statistical families remain separate. Detailed configurations, invalidated-run history, software versions, and cost diagnostics accompany the artifact and supplementary material.

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